PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 7

From series to shapes

Phase space, linear algebra and decompositions

Mathew Smith & Aneta Stefanovska

School of Physics and Astronomy | Lancaster University

Where we left off

From earlier lectures

- Limit cycle: a self-sustained oscillation (L2)
- Eigenvalues and eigenvectors (second-year linear algebra)
- The Jacobian matrix of a system (L3)
- Autocorrelation; the order carries the information (L6)

Today

A Phase space: the third view of a time series; fixed points, limit cycles, eigenvalues

B Decomposing data: correlation, SVD, PCA

C Outlook: other decompositions

Lecture 6: two views of a time series, in time and in frequency. Today the third: phase space.

Each decomposition assumes something about the shape of the set of points: look at the points first. Running example: the van der Pol oscillator of Lectures 2 and 6, $\mu = 2$.

PART A

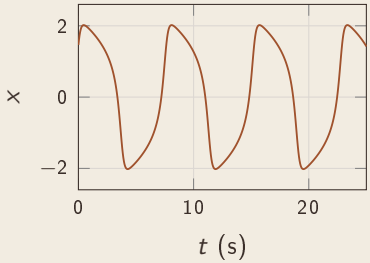
Phase space

A series as a path through the states of the system.

One series, three views

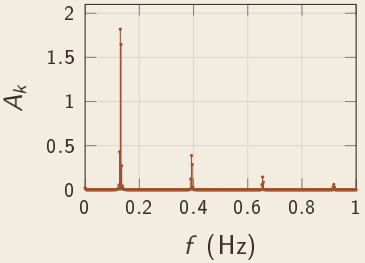
Problem: a plot of $x(t)$ shows one value at a time. How do we see the state of the system?

time domain (L6)



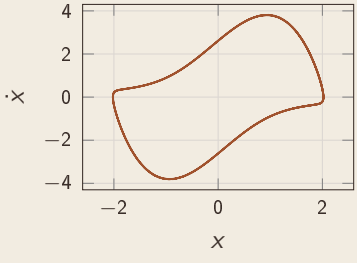
period, shape, slow and fast parts

frequency domain (L6)



$f_0 = 1/P$ and its harmonics

phase space (today)



the state: one closed curve

$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$. Right: each moment plotted as velocity $\dot{x}$ against position x – every period traces the same closed curve.

Data: van der Pol $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$, $\mu = 2$, RK4 (L1, L2), every 0.05 s; used throughout, noise added where stated.

State, phase space and trajectory

Problem: what do we need to know at one moment to predict the future?

- **State:** the numbers that, with the equations, fix the future (for a deterministic system; with noise, only its probabilities). For a second-order equation such as van der Pol: position and velocity, $(x, \dot{x})$.
- **Phase space:** the space of all possible states; here the plane $(x, \dot{x})$.
- **Trajectory:** the path of the state in time, starting from the initial condition (Lecture 1: an initial-value problem).

Autonomous and nonautonomous

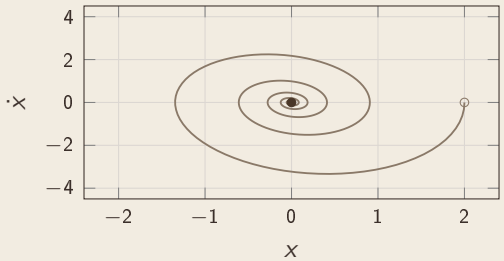
Autonomous system (L2): through each point of phase space passes exactly one trajectory – trajectories never cross.

Nonautonomous system: the same phase space, but the rule changes in time – the same state can lead to different futures. With slowly changing parameters, the fixed points and cycles of the frozen system move (slide 9).

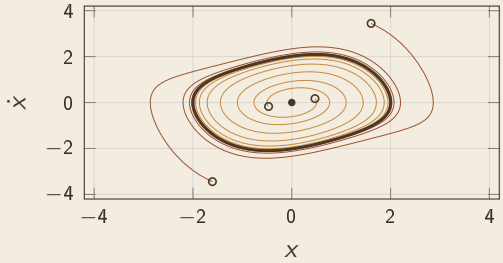
Fixed points and limit cycles

Problem: where do trajectories go in the long run?

Mass–spring–damper (L2): to a fixed point



van der Pol, $\mu = 0.3$: to a limit cycle



Fixed point: a state that does not change, $\dot{x} = 0$: for $(x, \dot{x})$ both $\dot{x} = 0$ and $\ddot{x} = 0$. **Limit cycle**: an isolated closed trajectory. **Attractor**: the set of states that trajectories approach in the long run – left a *point attractor*, right a *cycle attractor*. Open circles: starts. Right: from any start except the fixed point at the origin, the trajectory ends on the same cycle; its size and shape are set by the system. Larger μ: a distorted cycle ($\mu = 2$, slide 4), the same behaviour.

Data: simulated with RK4: mass–spring–damper ($c = 0.5$, $k = 4$, $m = 1$) and van der Pol ($\mu = 0.3$).

Reminder: eigenvalues and eigenvectors

Problem: for a linear system $\dot{\mathrm{x}} = A\mathrm{x}$, how do we know whether it decays, grows or oscillates without solving it?

An **eigenvector** v of A is a direction that A only stretches: $A\mathrm{v} = \lambda\mathrm{v}$, with the **eigenvalue** λ (real or complex). Along it, $\dot{\mathrm{x}} = \lambda\mathrm{x}$, so

$$\boxed{\mathrm{x}(t) = \sum_i c_i\, e^{\lambda_i t}\, \mathrm{v}_i} \qquad (c_i \text{ from the initial condition}).$$

Eigenvalue $\lambda = \alpha + i\omega$	Motion along v
$\alpha < 0$	decays at rate $\lvert\alpha\rvert$
$\alpha > 0$	grows: the state leaves
$\omega \neq 0$	oscillates at angular frequency ω

A nonlinear system $\dot{\mathrm{x}} = \mathrm{f}(\mathrm{x})$ near a fixed point: replace A by the Jacobian $J_{ij} = \partial f_i / \partial x_j$ there (L3).

Eigenvalues at the two fixed points

Problem: do the eigenvalues predict the motion we saw on slide 6?

Mass–spring–damper, $c = 0.5$, $k = 4$

$\dot{x} = v$, $\dot{v} = -kx - cv$:

$$A = \begin{pmatrix} 0 & 1 \\ -4 & -0.5 \end{pmatrix}, \quad \lambda = -0.25 \pm 1.98\, i$$

Decays at rate $0.25\,\mathrm{s}^{-1}$, oscillates at $\omega = 1.98$ rad/s: exactly $\omega_d = \sqrt{k/m - (c/2m)^2}$ of L2. Negative real parts: the fixed point is an **attractor** (the inward spiral).

van der Pol at the origin

Jacobian at $(0, 0)$:

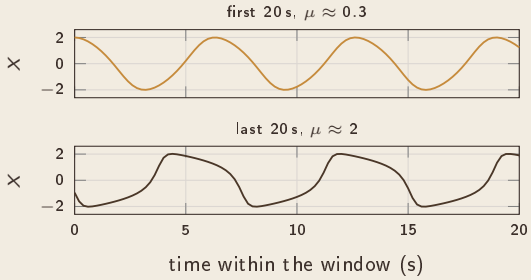
$$J = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}, \quad \lambda = \tfrac{1}{2}\left(\mu \pm \sqrt{\mu^2 - 4}\right)$$

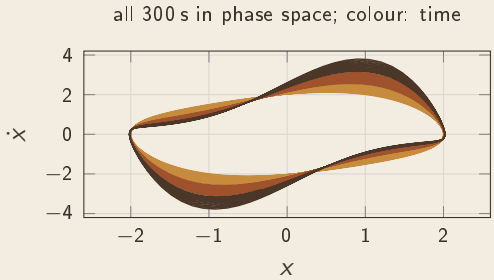
$\mu = 0.3$: $\lambda = 0.15 \pm 0.99\, i$, grows while oscillating (slide 6). $\mu = 2$: $\lambda = 1, 1$, leaves without oscillating. Positive real parts: the origin repels; the state settles on the limit cycle, as for $|x| > 1$ the damping turns positive.

Eigenvalues give the local motion near a fixed point; the limit cycle is a global, nonlinear property.

When the parameters change in time

Problem: what does phase space show for a nonautonomous system (L2)?





μ rises slowly from 0.3 to 2. Same phase space, but the rule changes: the cycle changes shape (largest $\dot{x}$ 2.5 → 3.8, period 6.4 → 7.6 s). Right (ochre: 0–100 s, red: 100–200 s, dark: 200–300 s): no single closed curve – states from different times are mixed.

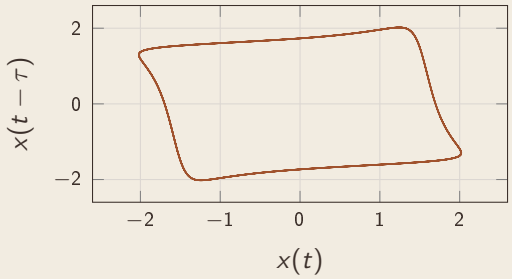
Phase space suits autonomous dynamics; nonautonomous change must be followed in time (L6).

Data: van der Pol with $\mu(t) = 0.3 + 1.7\,t/300\,\mathrm{s}$, RK4, 300 s.

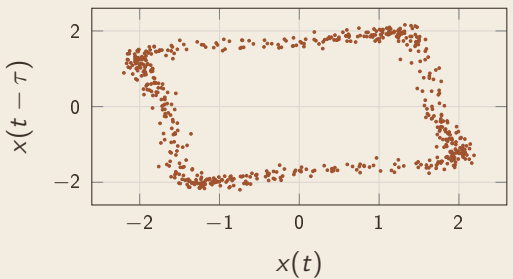
Phase space from one measured series: delay coordinates

Problem: usually only one quantity is measured, e.g. $x(t)$, and $\dot{x}$ from data is noisy (Lecture 4). Can we still draw the phase space?

without noise



with measurement noise, $\sigma = 0.1$

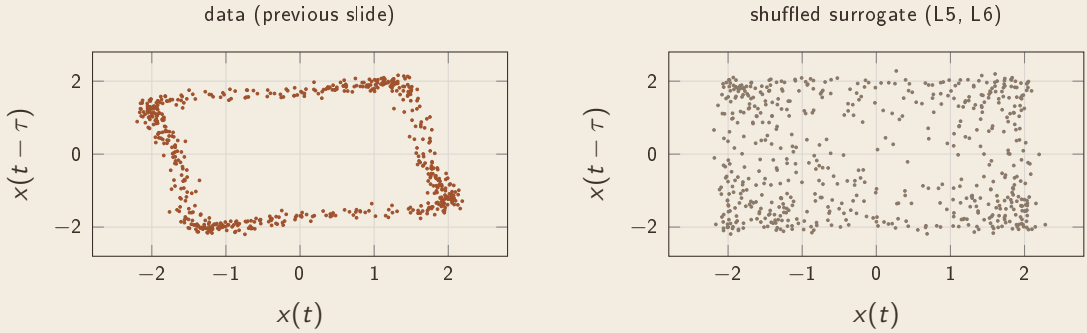


Delay coordinates: plot $x(t)$ against $x(t-\tau)$. A common starting point: τ where the autocorrelation (L6) first crosses zero, here $\tau = 1.95\,\mathrm{s} \approx P/4$; nearby τ should give much the same shape. The closed curve reappears, distorted. For more complex systems, use more delays: $(x(t), x(t-\tau), x(t-2\tau), \dots)$.

Data: van der Pol of slide 4, 200 s; right: plus independent Gaussian noise, $\sigma = 0.1$.

The shape is in the order

Problem: is the ring a property of the values, or of their order?



The same values, shuffled: same mean, s and histogram, but no ring. The shape in phase space is in the order of the values, as the rhythm was in Lecture 6. Noise blurs the ring; it does not remove it.

Data: the noisy series of slide 10; right: the same values in random order.

PART B

Decomposing data

How variables vary together, and along which directions: correlation, the singular value decomposition and principal components – for independent measurements (rows: people, samples, repeated experiments) and for time series (rows: successive moments, or delay vectors).

The algebra is the same for both; for a time series, neighbouring rows are not independent, and their order carries information.

Covariance and correlation

Problem: how do two variables x and y vary together, and how strongly? Same formulas for independent measurements and for time series.

Covariance:

$$s_{xy} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})$$

positive when both tend to be above (or below) their means together; units of x times units of y.

Correlation (Pearson): $r = s_{xy}/(s_x s_y)$, between -1 and 1, no units. It measures how close the points lie to a *straight line*.

Rank correlation (Spearman), r_S: r of the ranks (L5): a *monotonic* relation; robust to bad values.

Many variables: the covariance matrix

For m variables (the columns of X): $C_{jk} = s_{x_j x_k}$; on the diagonal, the variances $s_{x_j}^2$. With centred columns, $C = X^{\mathsf{T}}X/(N-1)$.

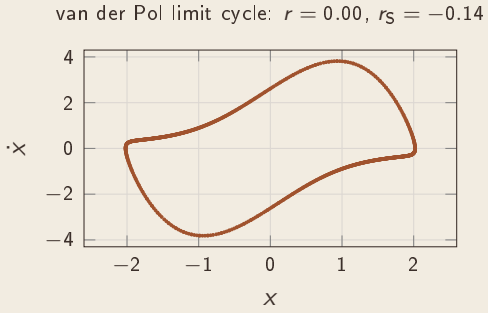
Example, noisy van der Pol (slide 10), x_n, x_{n-4} (0.2 s apart):

$$C = \begin{pmatrix} 2.22 & 2.16 \\ 2.16 & 2.22 \end{pmatrix}, \quad r = 0.976, \quad r_S = 0.965$$

Data: the noisy van der Pol series of slide 10.

Dependence versus association

Problem: if the correlation is zero, are the two variables unrelated?



For independent measurements the usual p-value of r holds. For a time series it does not: successive values are not independent, so it is far too small (L5).

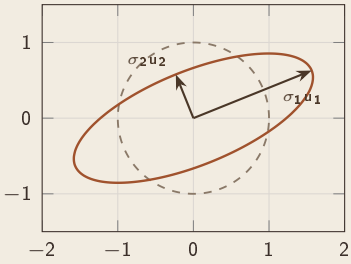
- **Association:** the two vary together. r measures straight-line association; r_S monotonic association.
- **Dependence:** knowing one changes what we know about the other. Here, given x, $\dot{x}$ can take only two values: complete dependence, zero correlation.
- A nonzero r does not show that one causes the other: both may follow a third variable, e.g. time.
- Nor does r measure **coupling**, the interaction in the dynamics (L8).

Data: van der Pol of slide 4, 200 s, without noise.

The singular value decomposition (SVD)

Problem: how do we find the main directions of a data table – a matrix of any shape, where eigenvectors do not apply?

A maps the unit circle to an ellipse



Every $m \times n$ matrix: $\boxed{A = U\Sigma V^{\mathsf{T}}}$

- V: orthonormal (perpendicular, unit-length) input directions $\mathbf{v}_i$
- Σ: **singular values** $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$, the stretches
- U: orthonormal output directions $\mathbf{u}_i$

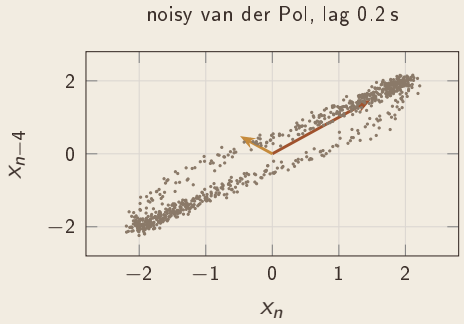
$A\mathbf{v}_i = \sigma_i \mathbf{u}_i$: a rotation, a stretch, a rotation.

What the σ_i tell:

- rank: the number of nonzero σ_i
- least squares: `lstsq` (L3) works through the SVD
- keep the k largest: the closest matrix of rank k (the step to PCA)

Principal component analysis (PCA)

Problem: many measured variables vary together. Along which directions do the data really vary?



Data matrix X: N rows (units, or moments of a series) $\times$ m columns (variables), each column minus its mean. Its SVD: $X = U\Sigma V^{\mathsf{T}}$

- **principal directions**: the v_k
- variance along v_k: $\sigma_k^2/(N-1)$
- **principal components**: the data in the new coordinates, XV

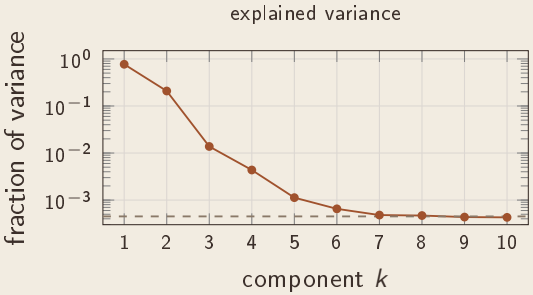
PCA ignores the row order: shuffled rows give the same result; for a series, order enters only via delay coordinates (next slide).

Left: pairs (x_n, x_{n-4}) of the noisy series of slide 10. Direction 1 (red, along the diagonal): 98.8% of the variance; direction 2 (ochre): 1.2%. Arrows: one standard deviation (direction 2 drawn 3$\times$ longer).

Data: the noisy van der Pol series of slide 10.

How many components? Ten delay coordinates

Problem: each row is now a 1.8-s piece of the record, 10 values 0.2 s apart. How many directions hold the signal?



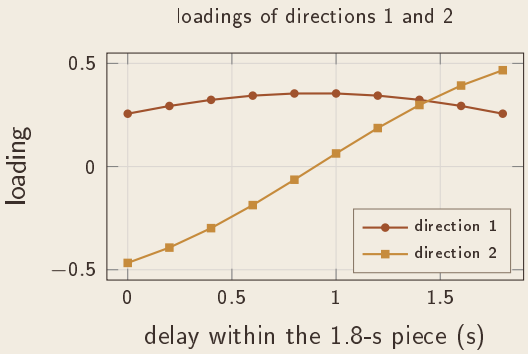


Components 1–2: 77% and 21%; 3–6: from 1.4% down to 0.07%; 7–10: a flat floor (dashed): independent noise adds the same variance, $\sigma^2 = 0.01$, to every direction. Above the floor: signal. With correlated noise the floor is not flat. Components 1–2 trace the limit cycle; 3–6 its non-sinusoidal shape (harmonics, L6). PCA of delay coordinates is called **singular spectrum analysis**.

Data: the noisy van der Pol series of slide 10, 4001 values.

Reading the loadings

Problem: what does each principal direction mean in terms of the original variables?



The **loadings** of direction k are the entries of v_k: the weight of each original variable in that component.

Direction 1: all weights of one sign, largest in the middle – an average over the piece: where the oscillator is.

Direction 2: weights rising steadily from -0.47 to $+0.47$ – late minus early: how fast it moves.

Read the pattern of the weights, not single values; the sign of a direction is arbitrary.

Data: as on the previous slide.

Standardising: the units decide the directions

Problem: variables in different units. Does PCA depend on the units we choose?

van der Pol state, 200 s: position x and velocity $\dot{x}$

Units	Direction 1	Variance in directions 1, 2
x in m, $\dot{x}$ in m/s	mostly x	50.3%, 49.7%
x in m, $\dot{x}$ in mm/s	$\dot{x}$ alone	100%, 0%
standardised	either: no preferred direction	50.0%, 50.0%

In mm/s the velocity numbers are 1000 times larger: their variance dominates, and PCA reports one component.

Standardise

$z_{ij} = (x_{ij} - \bar{x}_j)/s_j$: every column has mean 0 and standard deviation 1; PCA of z is PCA of the **correlation matrix**. Use it for different units or scales, not for delay coordinates (one quantity, one unit).

What PCA does and does not tell

Problem: PCA always returns directions. What do they mean?

PCA gives	**PCA does not give**
straight directions of largest variance	curved structure: the limit cycle, one closed curve, needs *two* components
the number of directions above the noise floor	independent causes: components are only uncorrelated
a lower-dimensional picture of the data	importance: a weak rhythm can have small variance
directions for any table of rows	the order of the rows: for a time series, only delay coordinates bring it in

Before reading any PCA: check the units (previous slide) and the noise floor (slide 17).

PART C

Outlook

Other decompositions, and the systems they suit.

Decompositions and the systems they suit

Problem: which decomposition suits which data?

Method	Idea	Assumes
PCA (today)	straight directions of largest variance	linear structure; one set of directions for the whole record
Singular spectrum analysis (today)	PCA of delay coordinates of one series	a stationary series
Dynamic mode decomposition	fit $x_{n+1} = Ax_n$ to many measured variables; each eigenvector of A is a mode with its own frequency and growth rate (slide 7)	linear, autonomous dynamics
Independent component analysis	unmix signals that are fixed mixtures of independent sources	independent, non-Gaussian sources; fixed mixing
Nonlinear mode decomposition	extract each oscillation from a time–frequency view (as the STFT, L6), keeping its harmonics in one mode; surrogates decide oscillation or noise	oscillations; frequency and amplitude may change in time
Clustering (L10)	groups of points: k-means, DBSCAN	round groups, or dense regions

Which assumption fails first?

Problem: the system types of Lectures 1 and 2 – which decompositions does each break?

System (L1, L2)	What goes wrong	Better suited
linear, autonomous, stationary (e.g. noise-driven)	little: the same straight directions throughout	PCA, SSA, DMD
linear, autonomous, decaying or growing	PCA mixes early and late	DMD (growth, decay rates)
nonlinear	a curve needs several straight directions; harmonics split into separate modes	phase space; NMD
nonautonomous	one fixed set of directions or frequencies averages the change	NMD; time-resolved views (L6)
stochastic, noisy	independent noise adds a floor to every component (slide 17)	components above the floor; surrogates
switching states	PCA mixes the states into one cloud	clustering (L10)

Classify the system first (L2): the classification tells which decomposition to trust.

Which method for which question?

Question	Method	Assumes	Check
What states does it visit?	phase portrait; delay coordinates	enough coordinates; good τ	shuffled surrogate; vary τ
Do two variables vary together?	r; rank r_S	straight line; monotonic	plot: $r = 0 \not\Rightarrow$ independent
Local motion at a fixed point	eigenvalues of J	close to the point	simulate
Main directions of variation	PCA (SVD of centred data)	straight directions; same units	noise floor; standardise; loadings

Before you decompose: four questions

1. What is the state of the system, and do I measure it, or must I reconstruct it? *(delay coordinates)*
2. Is the structure in the order of the values? *(compare with a shuffled surrogate)*
3. How many components stand above the noise floor – and is the structure straight or curved? *(PCA)*
4. Does my measure of association fit the relation? *($r = 0$ does not mean independent: plot the pair)*

Plot the points first; then choose the decomposition.

TO TAKE AWAY

In phase space a rhythm is a closed curve.

Correlation and PCA describe straight-line structure.

They can be applied as they stand when the variation lies along straight directions that stay the same through the data (for a time series: a stationary system), with variables in the same units or standardised.

Otherwise they still return numbers, read differently: a curve (a limit cycle) needs several straight components, and r can be zero despite complete dependence; a change in time gives one average set of directions; switching states give one mixed cloud.